Contribution of PM2.5 and air quality to the expansion of West Nile virus in the USA

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Abstract

West Nile virus (WNV) poses a significant public health threat globally, characterized by its mosquito-borne transmission. Despite its impact, no approved vaccine exists against WNV. This study examines the relationship between WNV outbreaks and air quality indicators (PM2.5 and AQI) in California from 2016 to 2022. Using Generalized Additive Models (GAM) and Causal Deep Learning (CDL), we find a marginally significant link between PM2.5 levels and neuroinvasive cases, while AQI does not show a notable effect. Additionally, we apply Poisson Event Weighted Moving Average (PEWMA) and Seasonal Autoregressive Integrated Moving Average (SARIMA) for forecasting, identifying seasonal trends in WNV cases that align with higher $PM_{2.5}$ levels in warmer months. These results highlight the importance of incorporating air quality considerations into public health strategies for managing WNV transmission

Keywords

West nile virus, PM2.5, air quality, causal deep learning

JEL Classification

C53, I18, Q5

Introduction

West Nile virus (WNV) poses a significant global public health risk. As one of the Nationally Notifiable Infectious Diseases in the United States, it is known as a mosquito-borne flavivirus. Despite its prevalence, there is currently no approved vaccine to prevent WNV infection (Suthar, Diamond, & Gale, 2013). WNV infection can be asymptomatic or progress to severe conditions such as West Nile fever or West Nile meningitis ("Clinical Characteristics and Functional Outcomes of West Nile Fever", 2004). This virus was originally discovered in the West Nile River in Uganda in 1937 (Smithburn, Hughes, Burke, & Paul, 1940). It entered New York in the United States in 1999, reaching the Pacific region in 2003, contributing to high West Nile virus cases in the northeastern United States metropolitan areas in 2002 (El Adlouni, Beaulieu, Ouarda, Gosselin, & Saint-Hilaire, 2007). In 2007 alone, WNV caused 109 deaths in the United States (Bernard & Kramer, 2001). Over the last twenty years, Colorado has witnessed several occurrences of WNV outbreaks, notably a significant outbreak in 2023 alone. Nevertheless, the underlying cause of WNV's resurgence in Colorado remains unknown. Throughout 2023, Colorado experienced the worst outbreak of West Nile virus in the United States, with more than 600 infections and 50 deaths. West Nile virus can cause severe neurological damage and brain swelling in severely ill patients. This was the deadliest year since the mosquito-borne disease first arrived in the United States in 1999 (McCrimmon, 2024).

WNV virus typically spreads during warm periods, and studies have indicated that elevated temperatures are associated with increased distribution of mosquitoes carrying WNV (Skaff et al., 2020). Additionally, numerous studies have explored the influence of meteorological factors, including precipitation and dew point temperature, on the dissemination of WNV (Soverow, Wellenius, Fisman, & Mittleman, 2009), (Farooq et al., 2022), (Hahn et al., 2015). The global warming linked to climate change has further intensified the spread of WNV. Various studies have explored the influence of climate change on the dissemination of WNV (Ferraccioli et al., 2023), (Mencattelli et al., 2023), (Erazo et al., 2024). Meanwhile, numerous studies have been conducted to explore the impact of air pollution on the transmission of West Nile virus (WNV), including the effects of ozone on the spread of the virus and the dynamic prediction of infectious diseases(Berrang-Ford et al., 2021),(Rannala, 2002),(El-Nadry, Li, El-Askary, Awad, & Mostafa, 2019), including WNV, associated with PM10 and SO2 (Karpati et al., 2004),(Goldgewicht, 2007),(DeFelice, Barker, & Shaman, 2019). However, to the best of our knowledge, there have been no studies investigating the impact of PM2.5 and climate on the spread of the WNV.

The impact of PM2.5 (fine particulate matter) on human health has garnered increasing attention. Research indicates that long-term exposure to high concentrations of PM2.5 may lead to a decline in immune system function, raising the risk of infections, including West Nile Virus. PM2.5 can trigger inflammatory responses in the body, potentially causing either an exaggerated or suppressed immune reaction (Signorini, Pignatti, & Coccini, 2021), (Miyata & van Eeden, 2011), and adversely affect lung health (Yang, Li, & Tang, 2020), (Wei & Tang, 2018). Some studies also suggest that PM2.5 can impair the functionality of immune cells, reducing resistance to pathogens and making individuals more susceptible to viral and other infections (Wei & Tang, 2018). Furthermore, PM2.5 may influence immune cell function, diminishing the body's ability to fend off pathogens (Bao, Fan, Cui, Sheng, & Zhu, 2017). Consequently, individuals in high PM2.5 environments may indeed face an increased risk of infection, particularly from pathogens transmitted by mosquitoes.

While it is recognized that PM2.5 alone may not directly enhance the transmission of West Nile Virus without vectors or a causal link, this study emphasizes the effects of air quality and particulate matter on health outcomes. It's crucial to acknowledge that numerous other factors contribute to health risks, such as environmental pollutants, dietary habits, levels of physical activity, and other lifestyle choices (Kulick, Kaufman, & Sack, 2023). This paper specifically aims to assess the extent of the impact that air quality and particulate matter have on health, while also considering potential interactions with other lifestyle factors. Air pollution poses significant challenges to public health, yet it also presents opportunities for action. By improving air quality, we can reduce health risks related to pollution and foster policy innovation.

Epidemiological research on the health impacts of PM2.5 suggests that due to the diversity and multitude of emission sources(Chay & Greenstone, 2003), (Chen et al., 2021), it is challenging to ascertain its primary origins, typically comprising primary emissions from combustion sources and secondary particles from atmospheric chemical reactions(Cao, 2024c)(Feng, Gao, Liao, Zhou, & Wang, 2016),(Tucker, 2000). Airborne particulate matter (PM) is mostly cleared through mucociliary mechanisms; however, particles with aerodynamic diameters smaller than 2.5 micrometers (PM2.5) tend to remain in our lungs (Janssen, Fischer, Marra, Ameling, & Cassee, 2013), (Burnett et al., 2014) and (Cao, 2024a). Epidemiological studies have indicated that this retention increases the incidence and mortality rates of cardiovascular and respiratory diseases, as well as the development of diabetes and adverse birth outcomes (Evans et al., 2013), (Zanobetti, Dominici, Wang, & Schwartz, 2014).(Watterson, Sorensen, Martin, & Coulombe Jr, 2007). An increase in PM2.5 concentrations in the previous week still leads to an increase in infectious diseases and influenza incidence(Croft et al., 2019)(Kloog, Melly, Ridgway, Coull, & Schwartz, 2012).

While many mechanical models exist for predicting the transmission of infectious diseases, they often pose challenges in parameterization and frequently necessitate making assumptions about parameters before the modeling process(Marcantonio et al., 2015),(Keeling & Rohani, 2008),(Keeling & Rohani, 2008).The advent of Artificial intelligence (AI) addresses this issue (Lundberg et al., 2020)(Cao, Debnath, & Alvarez, 2024). Many epidemiological studies have adopted machine learning, including MNV, to investigate the influence of meteorological changes on it(Beeman, Morrison, Unnasch, & Unnasch, 2021),(Skaff et al., 2020),(Smith et al., 2020). The drawback of machine learning and artificial intelligence is their often intricate and challenging interpretability. These models are commonly referred to as black box models.

This study aims to fill this gap by analyzing the relationship between WNV outbreaks, air quality indicators (specifically $PM_{2.5}$ and AQI) from 2016 to 2022, with a focus on California. We employ advanced analytical methods, including Generalized Additive Models (GAM) and Causal Deep Learning (CDL), to explore the complex interactions among these variables. Additionally, we utilize Poisson Event Weighted Moving Average (PEWMA) and Seasonal Autoregressive Integrated Moving Average (SARIMA) to assess seasonality and forecast future WNV cases. Our findings aim to inform public health strategies that effectively address the intertwined challenges of air pollution and infectious disease transmission.

Data and methods

Our first dataset is from the Centers for Disease Control and Prevention (CDC). We used annual WNV data from 2016 to 2021 in the United States, which is classified as an arboviral disease. In addition, we collected annual AQI data and $PM_{2.5}$ data from the United States Environmental Protection Agency (US EPA), and we combined all three kinds of data into a single dataset.

We focus on the California region as a case study and gather our second and third datasets. The second dataset comprises weekly West Nile Virus (WNV) data from California (CA), also obtained from the CDC. This dataset provides year-to-date cumulative counts from 2017 to 2022, enabling us to analyze trends and prevalence of the disease more effectively. We then merged this weekly WNV data with daily $PM_{2.5}$ data from the US EPA.

We are rolling up the weekly data into monthly data for our third dataset, which will provide insights into seasonal variations in the virus, and future support public health decision-making.

Descriptive statistical analysis

Annual variations in US

Figure 1 and Figure 2 illustrate the levels of AQI and PM2.5 across different states from 2016 to 2021, respectively. We observe trends in the variations of AQI (Air Quality Index) and PM2.5 (Particulate Matter) over the years. Figure 3 shows the variation in neuroinvasive case counts across different states. Table 1 presents the statistical analysis. We find that overall, AQI and PM2.5 values exhibit fluctuations across all years, with AQI generally being higher and PM2.5 fluctuating within a lower range. Recent years show an increasing trend in the median and mean values of AQI and PM2.5, indicating a deterioration in air quality. The neuroinvasive case counts also show a fluctuating trend but exhibit an overall increase. These trends may suggest a relationship between environmental pollution and health impacts.
Figure 3 shows the smooth terms for s(PM2.5) and s(AQI): The smooth curves depict the effects of PM2.5 and AQI on neuroinvasive cases. The shape of the curve indicates whether the relationship between the variables and neuroinvasive cases is linear. Significant curvature in the curve suggests a nonlinear relationship. The upper and lower bounds represent the confidence intervals of the smooth curves, which help assess the model's uncertainty. Narrow confidence intervals indicate a more reliable estimate for the smooth term. We find that the effect of PM2.5 on neuroinvasive cases is complex and not purely linear. The direction of the curve's slope indicates whether an increase in PM2.5 concentration has a positive or negative impact on neuroinvasive cases. In contrast, the effect of AQI appears to be linear. Tables 2 and 3 is from California data, which present summary statistics for West Nile Virus (WNV) cumulative case counts and PM2.5 concentrations across different

years, categorized by week and month. The weekly statistics indicate that WNV cases peaked in 2020 with a mean of 82.29, while the mean in 2022 significantly decreased to 39.85, suggesting a mitigation of the outbreak.

Regarding PM2.5 concentrations, the range in 2020 was relatively broad, with a mean of 12.589; however, this dropped significantly to 8.430 in 2022, reflecting an improvement in air quality. The monthly statistics further confirm these trends, showing that both the median and mean for WNV in 2021 were higher, indicating a more severe outbreak that year, while cases decreased in 2022.

Overall, there may be a correlation between the incidence of WNV and PM2.5 concentrations, warranting further investigation into the interplay between public health and environmental factors.

Monthly and weekely variations in CA

Figures 4 to 7 focus specifically on the California region rather than the entire United States. The weekly cumulative counts of WNV from 2017 to 2022 indicate a significant surge beginning after week 21, with values remaining close to zero or very low prior to that point. A closer examination of the monthly data reveals that WNV counts begin to spike starting from week 7. Although we are analyzing cumulative counts—which naturally increase over time as more cases accumulate—there is still a clear upward trend observed from August through October. This highlights the seasonality and rising prevalence of WNV during this period in California.

In contrast to the WNV data, the $PM_{2.5}$ data is presented as weekly or monthly averages rather than cumulative totals. A detailed analysis of the weekly $PM_{2.5}$ data reveals a rise accompanied by significant fluctuations beginning in week 33, which coincides with the weekly trends observed in WNV cases. When examining the monthly $PM_{2.5}$ concentrations, we notice a pronounced increase from July to October, indicating heightened pollution levels during this period. Interestingly, WNV cases exhibit a similar upward trend during these months, which align with California's typically hot summer season. This correlation suggests a possible seasonal relationship between WNV infections and $PM_{2.5}$ concentrations, pointing to environmental factors that may influence both public health and air quality during these warmer months.

This suggests a potential seasonal association between West Nile virus (WNV) infection cases and $PM_{2.5}$ concentrations during California's hot summer season. This association may be related to environmental factors, suggesting that air pollution may affect public health during specific periods. Therefore, monitoring $PM_{2.5}$ levels may help understand and prevent the spread of WNV.

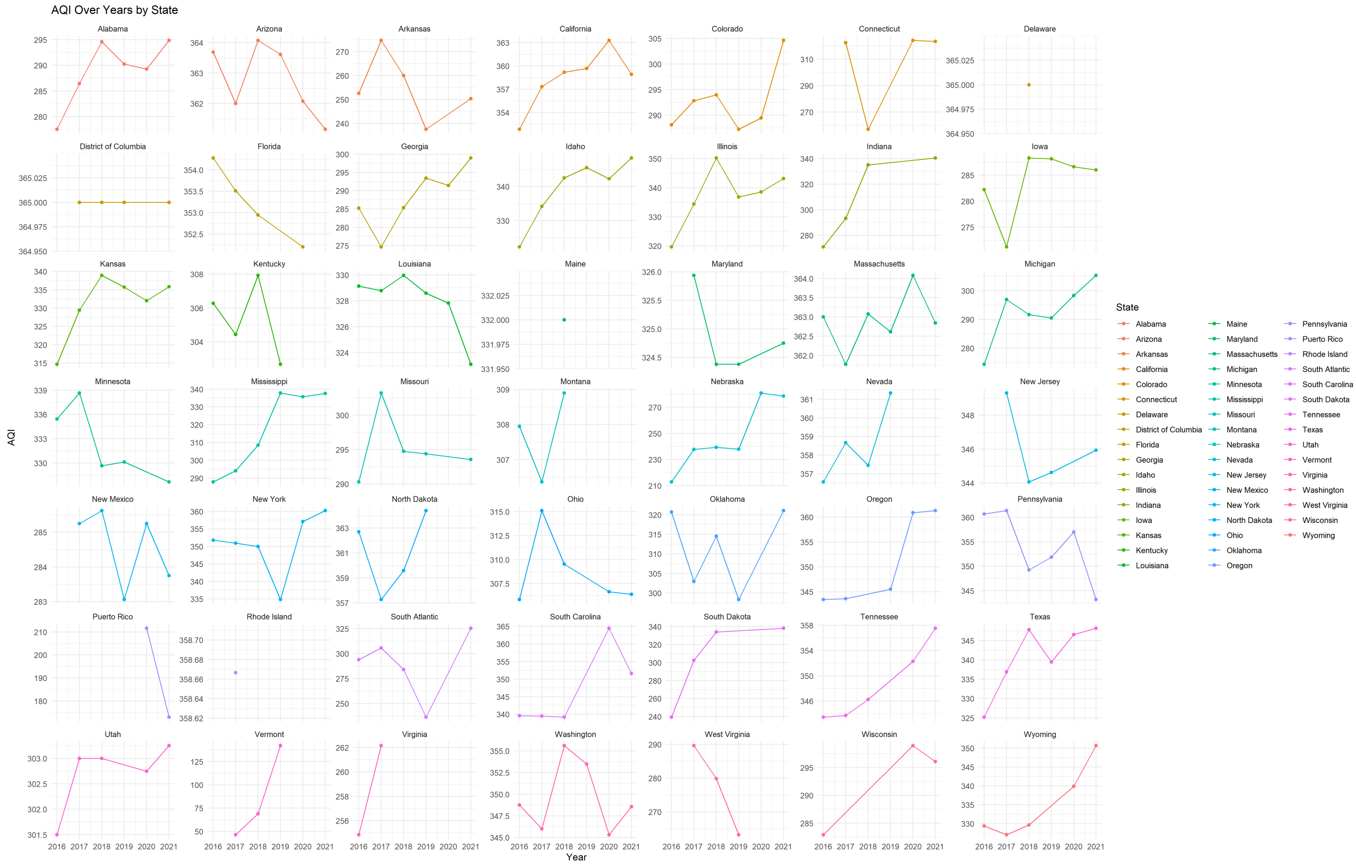


Figure 1: AQI Over Years by State

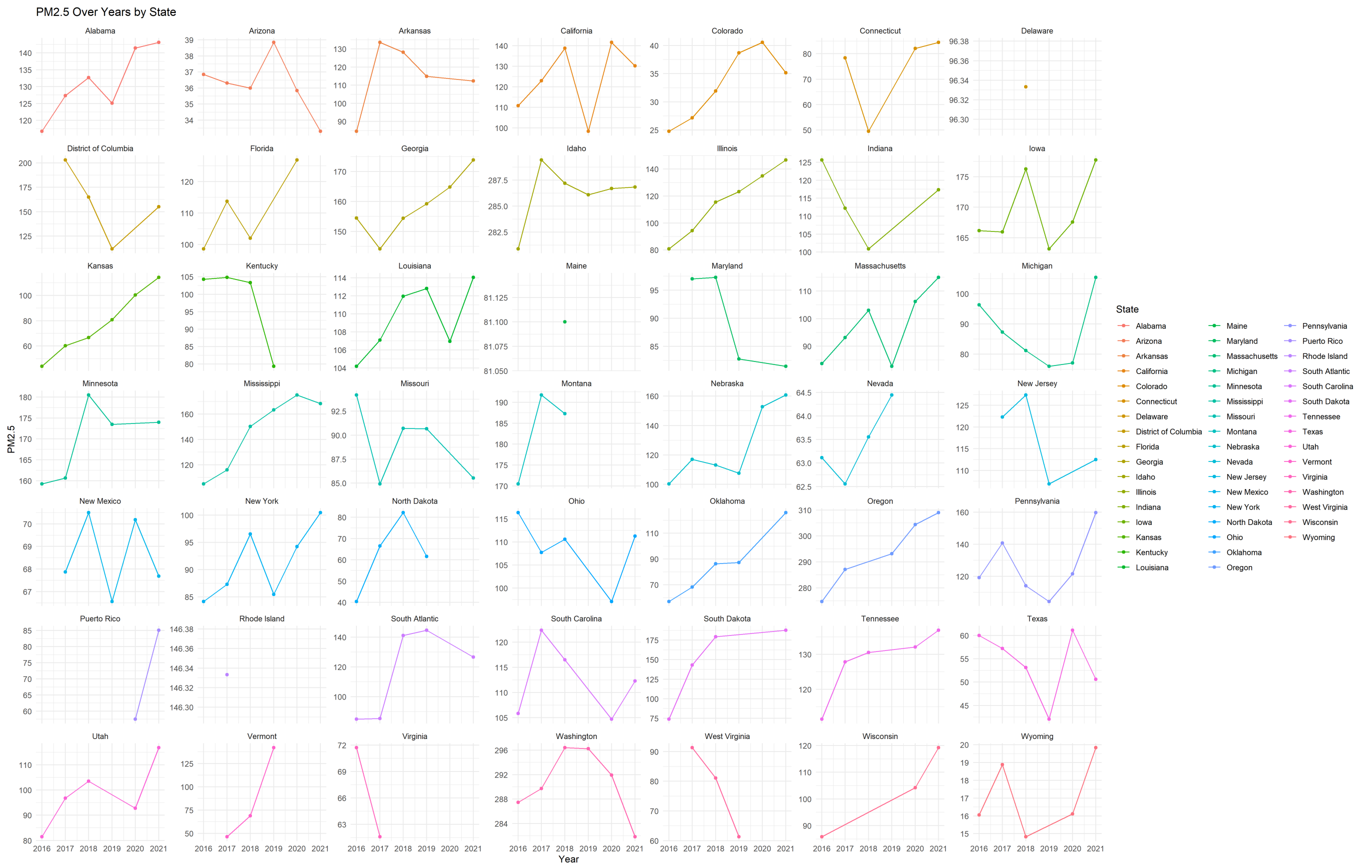


Figure 2: PM2.5 Over Years by State

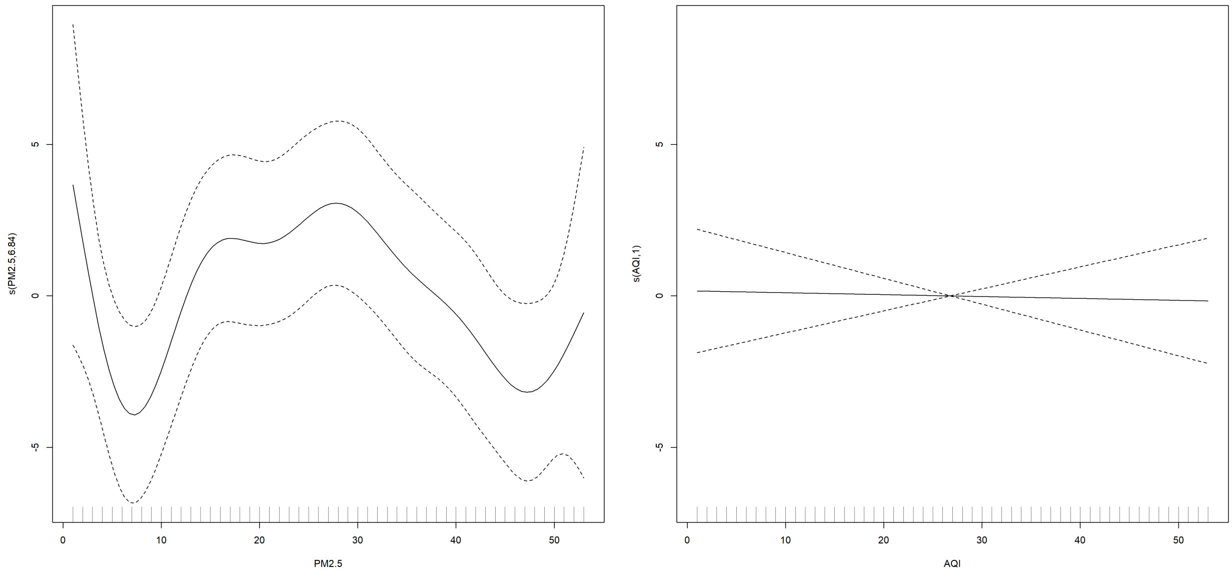
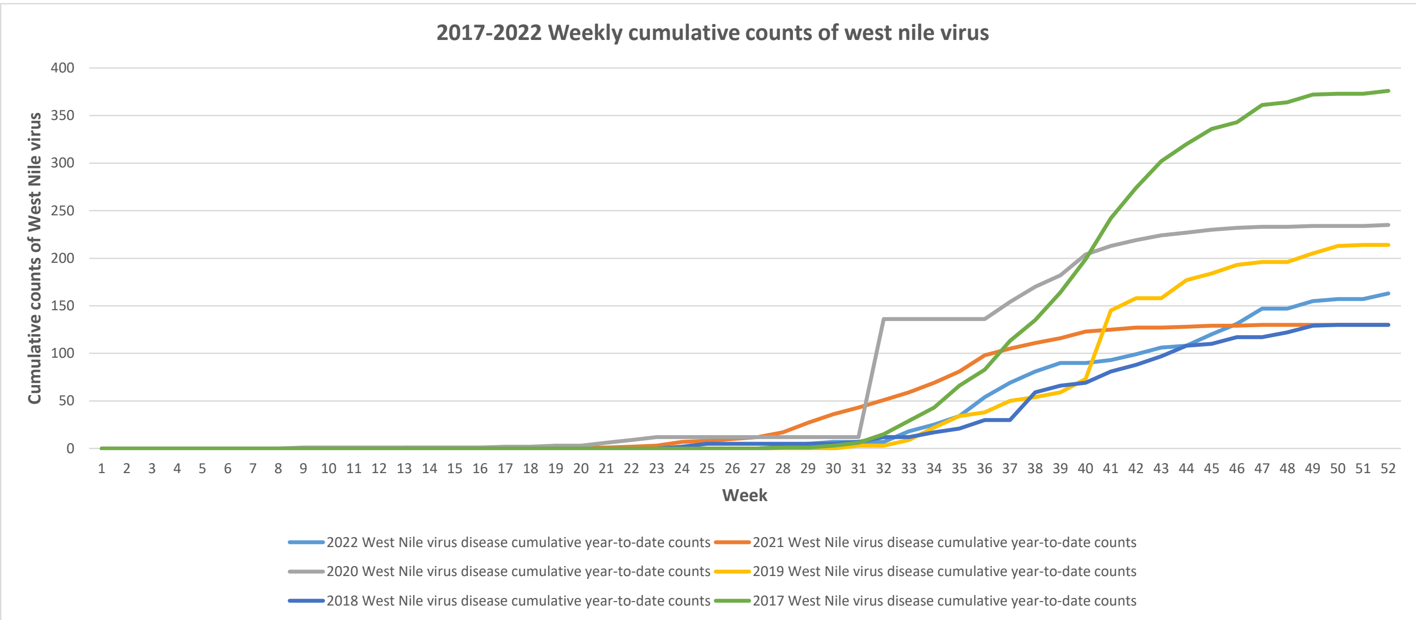


Figure 3: Neuroinvasive cases over years by state

Figure 4



2017-2022 Weekly cumulative counts of west nile virus
Cumulative counts of West Nile virus
0
50
100
150
200
250
300
350
400
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
Week
2022 West Nile virus disease cumulative year-to-date counts
2021 West Nile virus disease cumulative year-to-date counts
2020 West Nile virus disease cumulative year-to-date counts
2019 West Nile virus disease cumulative year-to-date counts
2018 West Nile virus disease cumulative year-to-date counts
2017 West Nile virus disease cumulative year-to-date counts

Figure 5

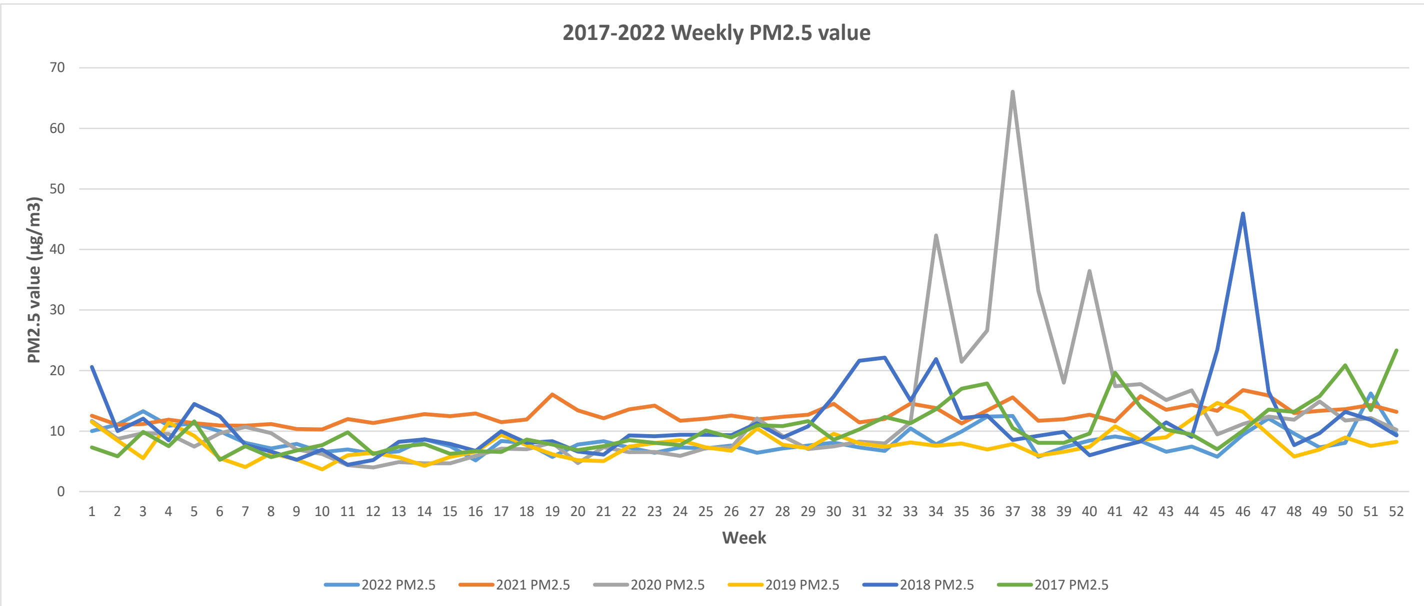

2017-2022 Weekly PM2.5 value
PM2.5 value (µg/m3)
70
60
50
40
30
20
10
0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
Week
2022 PM2.5
2021 PM2.5
2020 PM2.5
2019 PM2.5
2018 PM2.5
2017 PM2.5

Figure 6

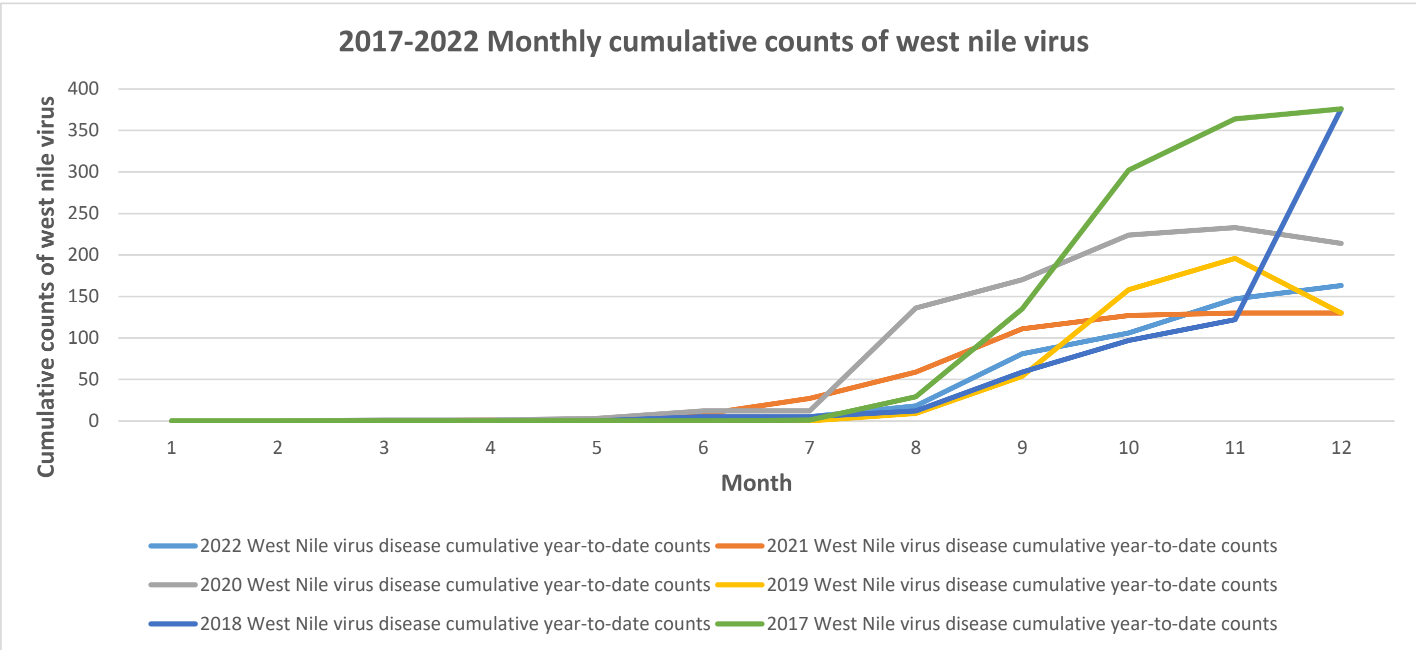

2017-2022 Monthly cumulative counts of west nile virus
Cumulative counts of west nile virus
400
350
300
250
200
150
100
50
0
1
2
3
4
5
6
7
8
9
10
11
12
Month
2022 West Nile virus disease cumulative year-to-date counts
2021 West Nile virus disease cumulative year-to-date counts
2020 West Nile virus disease cumulative year-to-date counts
2019 West Nile virus disease cumulative year-to-date counts
2018 West Nile virus disease cumulative year-to-date counts
2017 West Nile virus disease cumulative year-to-date counts

Figure 7

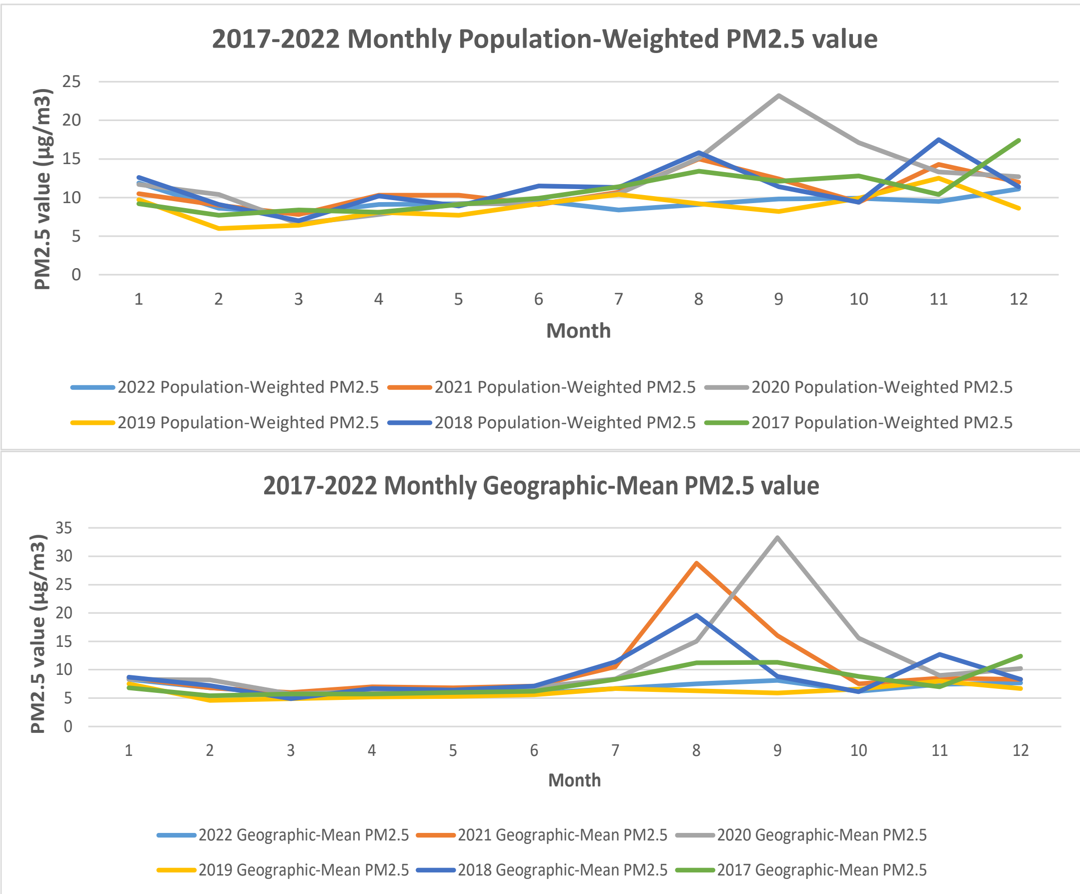

2017-2022 Monthly Population-Weighted PM2.5 value
PM2.5 value (µg/m3)
25
20
15
10
5
0
1
2
3
4
5
6
7
8
9
10
11
12
Month
2022 Population-Weighted PM2.5
2021 Population-Weighted PM2.5
2020 Population-Weighted PM2.5
2019 Population-Weighted PM2.5
2018 Population-Weighted PM2.5
2017 Population-Weighted PM2.5
2017-2022 Monthly Geographic-Mean PM2.5 value
PM2.5 value (µg/m3)
35
30
25
20
15
10
5
0
1
2
3
4
5
6
7
8
9
10
11
12
Month
2022 Geographic-Mean PM2.5
2021 Geographic-Mean PM2.5
2020 Geographic-Mean PM2.5
2019 Geographic-Mean PM2.5
2018 Geographic-Mean PM2.5
2017 Geographic-Mean PM2.5

Table 1: Statistical Analysis Table

	Neuroinvasive_2016	Non_neuroinvasive_2016	Neuroinvasive_2017	Non_neuroinvasive_2017	Neuroinvasive_2018	Non_neuroinvasive_2018
Range	1-30	1-21	1-39	1-26	1-43	1-31
Median	14.00	9.00	16.00	10.00	19.00	13.00
Mean	14.09	7.98	16.75	10.66	19.53	13.89
	Neuroinvasive_2019	Non_neuroinvasive_2019	Neuroinvasive_2020	Non_neuroinvasive_2020	Neuroinvasive_2021	Non_neuroinvasive_2021
Range	1-22	1-17	1-22	1-17	1-32	1-17
Median	9.00	5.00	8.00	2.00	15.00	5.00
Mean	10.02	5.75	9.02	5.21	15.04	5.21
	AQI_2016	PM2.5_2016	AQI_2017	PM2.5_2017	AQI_2018	PM2.5_2018
Range	57.5-366.0	16.06-287.45	46.5-365.0	18.88-289.73	69.0-365.0	14.83-296.40
Median	317.6	96.53	323.3	104.82	329.7	103.50
Mean	309.3	107.83	313.3	114.81	314.8	119.55
	AQI_2019	PM2.5_2019	AQI_2020	PM2.5_2020	AQI_2021	PM2.5_2021
Range	142.0-365.0	6.61-296.26	81.67-366.0	16.11-304.39	82.0-365.0	19.83-309.00
Median	330.1	103.00	331.21	105.00	327.7	114.18
Mean	316.1	114.80	319.19	119.22	320.0	127.87

Table 2: Summary Statistics for WNV Counts and PM2.5 by Week

Measure	WNV 2022 Cum Counts	2022 PM2.5	WNV 2021 Cum Counts	2021 PM2.5	WNV 2020 Cum Counts	2020 PM2.5
Range	0.00 - 163.00	5.160 - 16.202	0.00 - 130.00	10.26 - 16.77	0.00 - 235.00	3.984 - 66.072
Median	0.00	7.841	11.00	12.50	12.00	9.483
Mean	39.85	8.430	48.69	12.74	82.29	12.589
Measure	WNV 2019 Cum Counts	2019 PM2.5	WNV 2018 Cum Counts	2018 PM2.5	WNV 2017 Cum Counts	2017 PM2.5
Range	0.00 - 214.00	3.671 - 14.633	0.00 - 130.00	4.450 - 45.934	0.00 - 376.00	5.267 - 23.328
Median	0.00	7.489	5.00	9.295	0.00	9.181
Mean	49.96	7.709	32.96	11.357	94.12	10.210

Table 3: Summary Statistics for WNV and PM2.5 by Month

Measure	WNV 2022 Cum Counts	PM2.5 Wgt 2022	PM2.5 Geo 2022	WNV 2021 Cum Counts	PM2.5 Wgt 2021	PM2.5 Geo 2021
Range	0.00 - 163.00	7.900 - 28.800	5.700 - 11.900	0.00 - 130.00	7.800 - 15.000	6.000 - 28.800
Median	2.50	9.508	6.750	17.50	10.400	7.900
Mean	43.33	9.508	6.858	49.50	10.900	10.158
Measure	WNV 2020 Cum Counts	PM2.5 Wgt 2020	PM2.5 Geo 2020	WNV 2019 Cum Counts	PM2.5 Wgt 2019	PM2.5 Geo 2019
Range	0.00 - 233.00	6.60 - 23.20	5.700 - 8.300	0.00 - 196.00	6.000 - 12.500	4.600 - 8.000
Median	12.00	11.10	8.350	5.00	8.900	6.100
Mean	83.83	12.23	11.075	56.33	8.825	6.108
Measure	WNV 2018 Cum Counts	PM2.5 Wgt 2018	PM2.5 Geo 2018	WNV 2017 Cum Counts	PM2.5 Wgt 2017	PM2.5 Geo 2017
Range	0.00 - 376.00	7.000 - 17.400	4.900 - 19.600	0.00 - 376.00	7.700 - 17.400	5.400 - 12.400
Median	5.00	11.350	7.750	0.50	10.150	6.900
Mean	56.33	11.342	8.992	100.6	10.825	7.900

Methods

The empirical approach employed in this study is as follows:

$$\mathrm{WNV}_t = f(P_t, A_{t,w}) \tag{1}$$

Here, WNV_t represents the outcome of interest, while $A_{t,w}$ denotes the air quality index , with the subscript w indicating AQI values from 1 to w. The subscript t refers to time, ranging from 1 to T and measured in years. P_t represents $\mathrm{PM}_{2.5}$, also measured in years.

State fixed effects are incorporated to assess how fluctuations in AQI and pollution levels within states influence changes. To address seasonal variations, the second empirical approach is:

$$\mathrm{WNV}_m = f(P_m) \tag{2}$$

In this equation, WNV_m again denotes the outcome of interest, with the subscript m indicating time, ranging from 1 to m. P_m represents $\mathrm{PM}_{2.5}$, measured in weeks and months. We employ the generalized additive model and causal deep learning for the first approach, while the second approach utilizes a poisson event-weighted moving average and seasonal autoregressive integrated moving average.

Generalized additive model (GAM)

The generalized additive model (GAM) exhibits unique advantages in the study of infectious diseases. It effectively explores complex nonlinear relationships within disease transmission processes, such as the interaction between disease transmission rates and environmental factors, as well as population behaviors(Jbilou & El Adlouni, 2012). By using smooth functions, the GAM model can accurately capture the dynamic changes in virus or bacteria transmission, thereby providing robust predictions and policy recommendations. This flexibility makes the GAM model a powerful tool for investigating the mechanisms and impacts of infectious disease spread, contributing to the development of more effective strategies for disease prevention and public health policies(Scrucca, 2022).

Causal deep learning (CDL)

Causal deep learning (CDL) offers several advantages for policy evaluation by combining the strengths of causal inference and deep learning techniques(Whata & Chimedza, 2022). CDL can handle large and complex datasets, including high-dimensional features and unstructured data like text, images, and time series, which are common in policy evaluations. It automatically learns representations from raw data, capturing intricate patterns and relationships that traditional methods might miss(Singla, Wallace, Triantafillou, & Batmanghelich, 2021). Additionally, CDL effectively captures non-linear relationships, providing more accurate and nuanced insights into the impact of policies. This flexibility allows for a deeper understanding of policy effects, leading to better-informed decisions and more effective policy designs.

CDL offers significant benefits for public health by enhancing the ability to evaluate and understand the impact of health policies and interventions. CDL can handle vast and complex datasets, including electronic health records, genomic data, and social media activity, allowing for a comprehensive analysis of public health issues. It captures intricate patterns and non-linear relationships that traditional methods might overlook, providing more accurate insights into how different factors influence health outcomes. This leads to a better understanding of the effectiveness of health policies, targeted interventions, and resource allocation, ultimately improving health outcomes and promoting more efficient and effective public health strategies(Prosperi et al., 2020).

Poisson event weighted moving average (PEWMA)

We use poisson event weighted moving average (PEWMA) is a statistical method that combines the Poisson distribution with weighted moving averages, primarily used for modeling and forecasting time series data, especially for count data such as event occurrences(Brandt, Williams, Fordham, & Pollins, 2000). This method assigns specific weights to observations within a given time window based on the Poisson distribution, allowing more recent observations to have a greater influence on the current prediction. This effectively smooths the time series data, reducing the impact of short-term fluctuations while highlighting trends and periodic changes. The flexibility of the Poisson Weighted Moving Average allows for the adjustment of weight parameters to fit different data characteristics, making it widely applicable in fields such as public health, traffic management, and meteorological monitoring, thereby providing an effective tool for modeling and forecasting count data.

Seasonal autoregressive integrated moving average (SARIMA)

We also chose seasonal autoregressive integrated moving average (SARIMA)(Cao, 2024b), a statistical model used for time series forecasting that is particularly well-suited for data with seasonal characteristics. Its basic form is SARIMA(p, d, q)(P, D, Q)[s], where p, d, and q represent the number of non-seasonal autoregressive, differencing, and moving average terms, respectively, while P, D, and Q correspond to the seasonal components, with s indicating the seasonal period. By conducting stationarity tests, parameter identification, and model fitting, SARIMA effectively captures trends and seasonal variations in the data, enabling accurate predictions. Although it is widely used in various fields such as meteorology, economics, and traffic flow analysis, it may have limitations when dealing with complex seasonal patterns.

Results

Exploring the relationship between PM2.5 Exposure, AQI, and WNV Outcomes

We initially employed a gaussian GAM model with an identity link function to investigate the causal relationship between PM2.5 exposure, AQI, and health outcomes. Subsequently, we visualized the results from the GAM analysis. Our findings indicate that the smoothed terms $s(\text{PM2.5})$ and $s(\text{AQI})$ do not exhibit a causal relationship with variations in Neuroinvasive cases.

Specifically, the GAM analysis enabled us to model the non-linear effects of PM2.5 and AQI on the outcome of interesting, Neuroinvasive cases. Despite exploring these relationships graphically and statistically, we did not observe evidence supporting a causal link between PM2.5 or AQI levels and changes in Neuroinvasive cases. This suggests that other factors not captured by our models may influence neuroinvasive outcomes, highlighting the complexity of environmental and health interactions.

Formula:

$$\text{Neuroinvasive} \sim s(\text{PM2.5}) + s(\text{AQI}) + \text{factor(year)}$$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> |t|)
(Intercept)	31.058	34.792	0.893	0.3730
factor(year)2017	36.936	47.148	0.783	0.4342
factor(year)2018	50.511	47.404	1.066	0.2878
factor(year)2019	-9.789	50.935	-0.192	0.8478
factor(year)2020	21.626	52.979	0.408	0.6835
factor(year)2021	89.816	49.417	1.818	0.0705

Approximate significance of smooth terms:

	edf	Ref.df	F p-value
s(PM2.5)	1	1	2.707
s(AQI)	1	1	0.090

In this analysis, the dependent variable **Neuroinvasive** is modeled using a Gaussian family with an identity link function. The formula includes smoothed terms for **PM2.5** and **AQI**, as well as a categorical factor for **year**.

Regarding the parametric coefficients, none of the factors for different years (**factor(year)**) show statistically significant effects at the conventional levels (p > 0.05). However, the coefficient for **factor(year)2021** shows a trend towards significance (p = 0.0705).

The approximate significance tests for the smoothed terms (**s(PM2.5)** and **s(AQI)**) indicate that only the term **s(PM2.5)** shows a marginally significant effect (p = 0.101). In contrast, **s(AQI)** does not exhibit significant effects (p = 0.764).

These results suggest that while there may be some association between **PM2.5** levels and the neuroinvasive outcome, the relationships with **AQI** and year are not statistically significant based on this analysis.

Counterfactual policy causal analysis

We then employed CDL to investigate the impact of PM2.5 policy changes on Neuroinvasive disease cases. We initially conducted counterfactual policy analysis using linear models, assuming a 10% For policy simulation, we simulated a scenario where PM2.5 levels were reduced by 10%. Due to the limited size of our dataset, we implemented a simple sequential neural network with two hidden layers. This model was trained to predict outcomes based on simulated policy changes and was evaluated against actual data. The results were subsequently visualized to illustrate the model's predictions compared to real-world observations. This approach allowed us to assess the potential effects of PM2.5 reduction policies on Neuroinvasive disease cases, despite the challenges posed by our dataset's size.

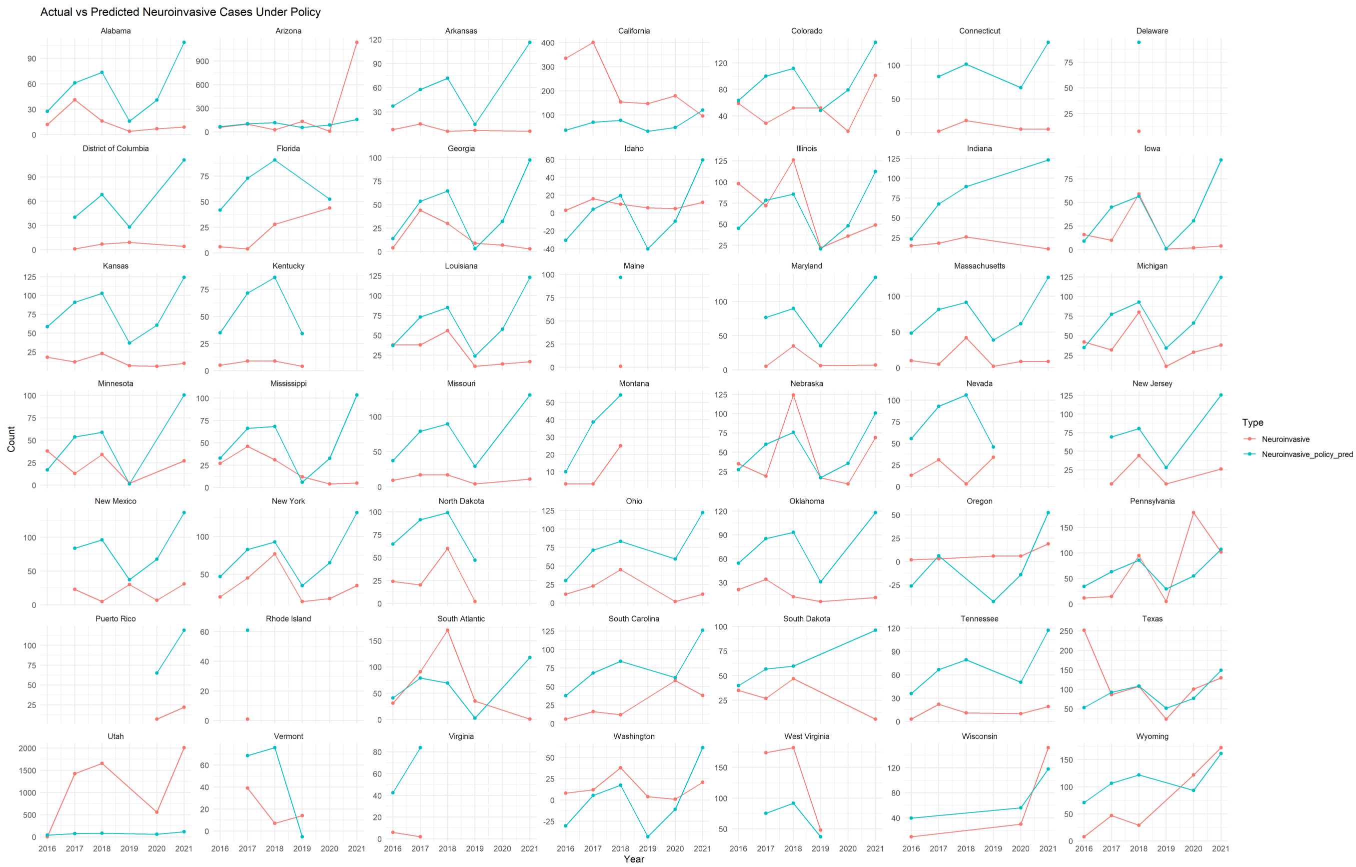


Figure 8: Actual vs Predicted Neuroinvasive Cases Under Policy

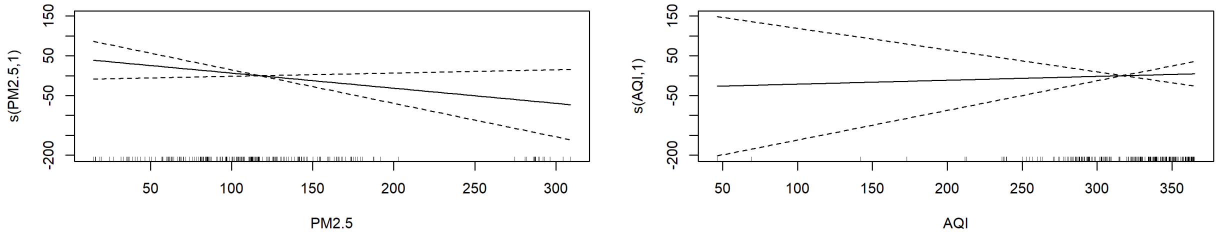


Figure 9: Visualize GAM model

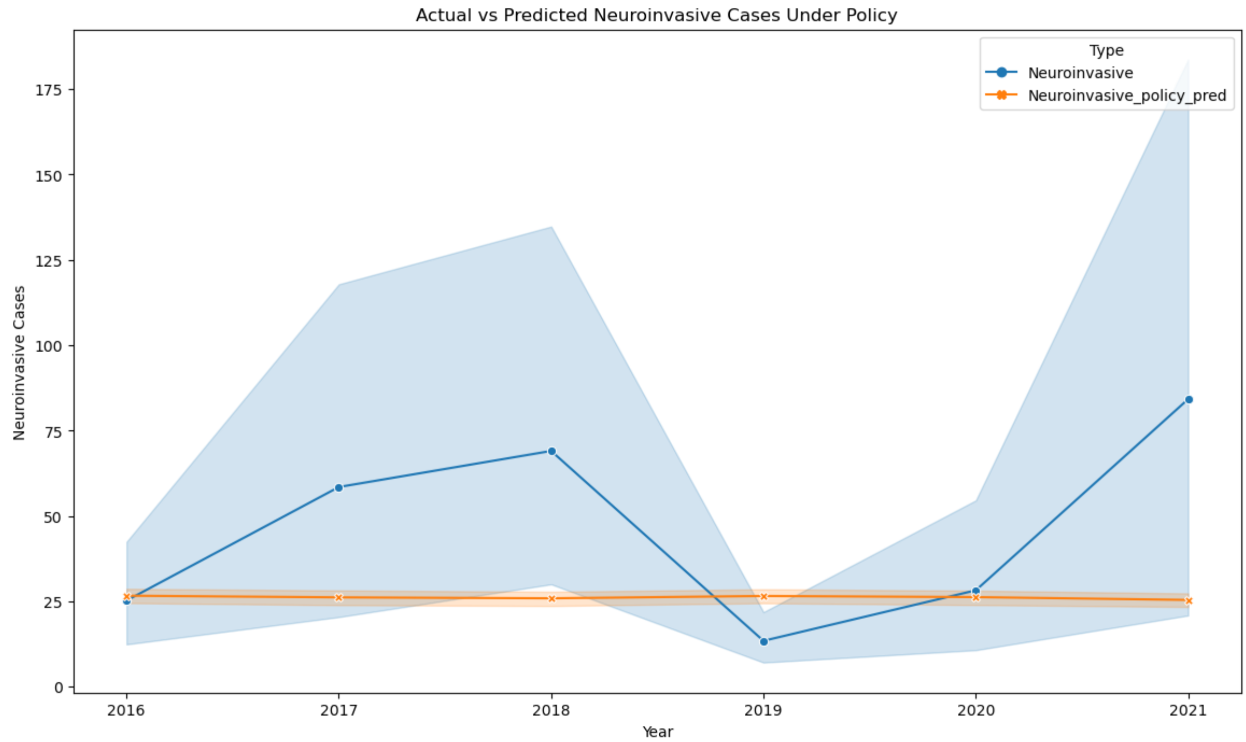


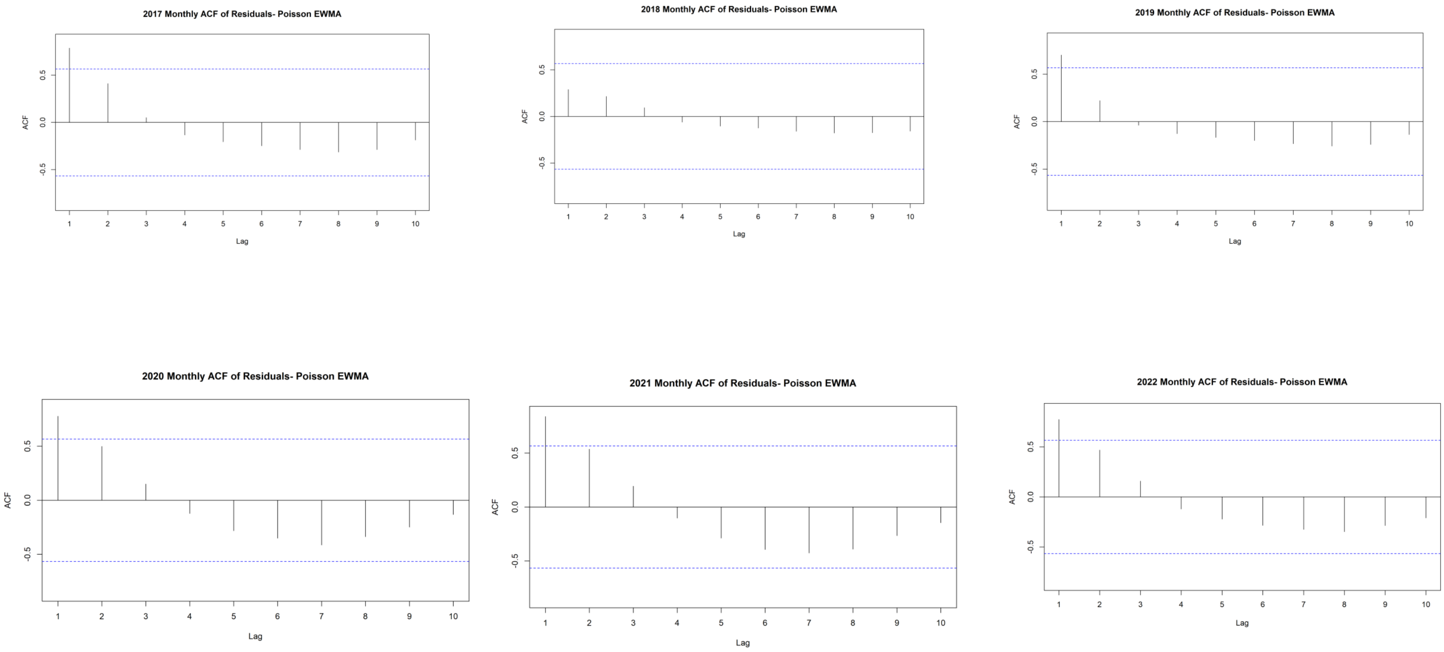
Figure 10: Result of causal deep learning

Seasonality exploration

Visualizing and exploring seasonality in time series data is essential for understanding underlying patterns and trends. We use PEWMA and SARIMA for seasonal exploration is highly beneficial. PEWMA smooths time series data to reduce noise and adjusts weights to give more significance to recent observations, enhancing adaptability. On the other hand, SARIMA is specifically designed to handle seasonal data, effectively capturing seasonal patterns while considering both autoregressive and moving average components, making it suitable for complex time series analysis. Combining these two methods allows for clearer identification of trends and seasonal variations on a smoothed dataset, thereby improving the accuracy of future data predictions. We use autocorrelation plots (ACF) to identify lag correlations, which can highlight periodic cycles within the data. If strong autocorrelation is observed at lag 12 for the monthly data, this indicates the presence of an annual seasonal pattern, as data points in the same month of the same year are correlated. Conversely, if the ACF decays rapidly, this may indicate a lack of seasonality or a more complex pattern in the data that requires further investigation. Figures 11 and 12 observe that the ACF plot shows a rapid decay after lag 12, with most lag values approaching zero. This suggests that the data may exhibit more complex patterns, such as trends or cyclic behavior than seasonality.

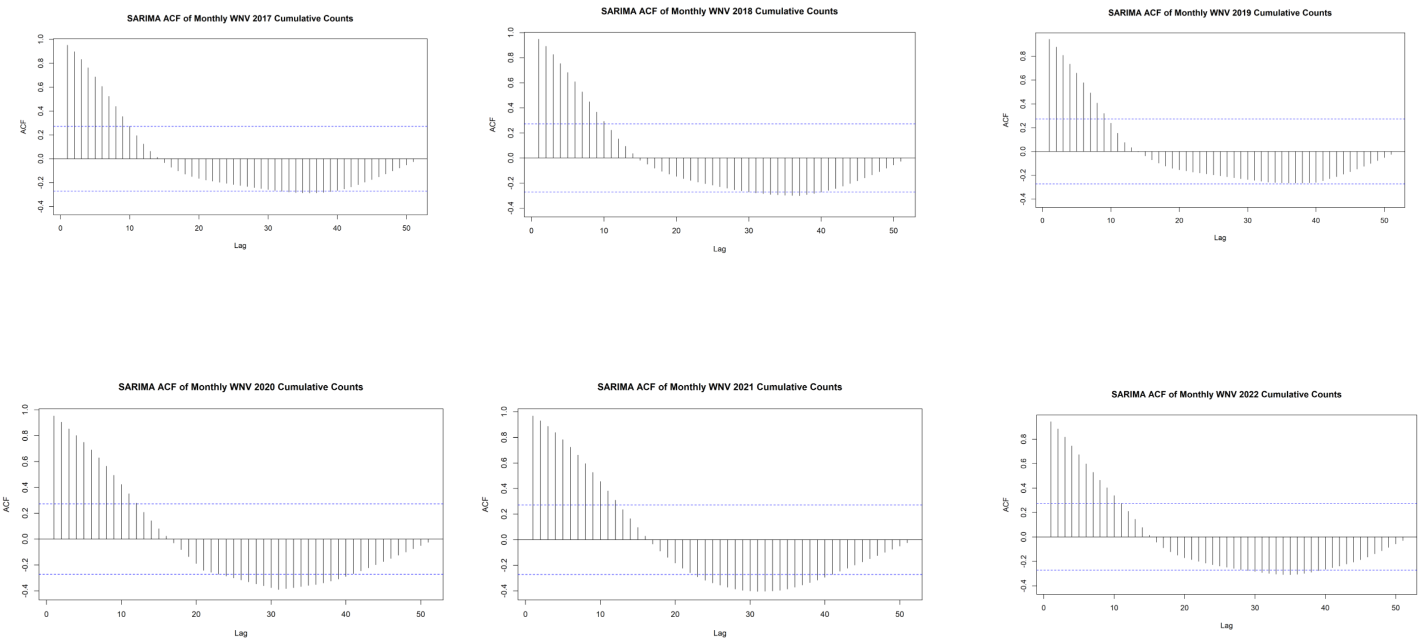
We chose to use PEWMA for forecasting instead of SARIMA primarily because PEWMA excels at handling data noise and short-term fluctuations. It effectively smooths time series data, highlighting recent trends, which is particularly valuable in scenarios that require quick responses. Additionally, the flexibility of PEWMA allows us to dynamically adjust weights based on actual data changes, enabling better capture of immediate variations. While SARIMA performs well in capturing long-term seasonal patterns, its model development and parameter tuning can be relatively complex, making it less suitable for rapid forecasting needs. Therefore, in certain situations, PEWMA can provide more intuitive and prompt forecasting results. We made predictions for the next 52 weeks (see Figure 13) and for the upcoming 12 months (see Figure 14). The red curve represents the predicted values, while the blue curve indicates the actual values. Figure 13 indicates that the virus data for the next year will change almost in parallel; note that this refers to cumulative WNV cases rather than weekly variations. For instance, the prediction for the cumulative WNV cases over the next 52 weeks in 2017 was about 150 cases, while the actual number in 2018 exceeded 150 cases. Similarly, the 2021 prediction for 2022 was approximately 170 cases, but the actual number in 2022 was below 150 cases.The same analysis is applied to monthly forecasts for the next 12 months in figure 14, and we find that there are deviations.

Figure 11



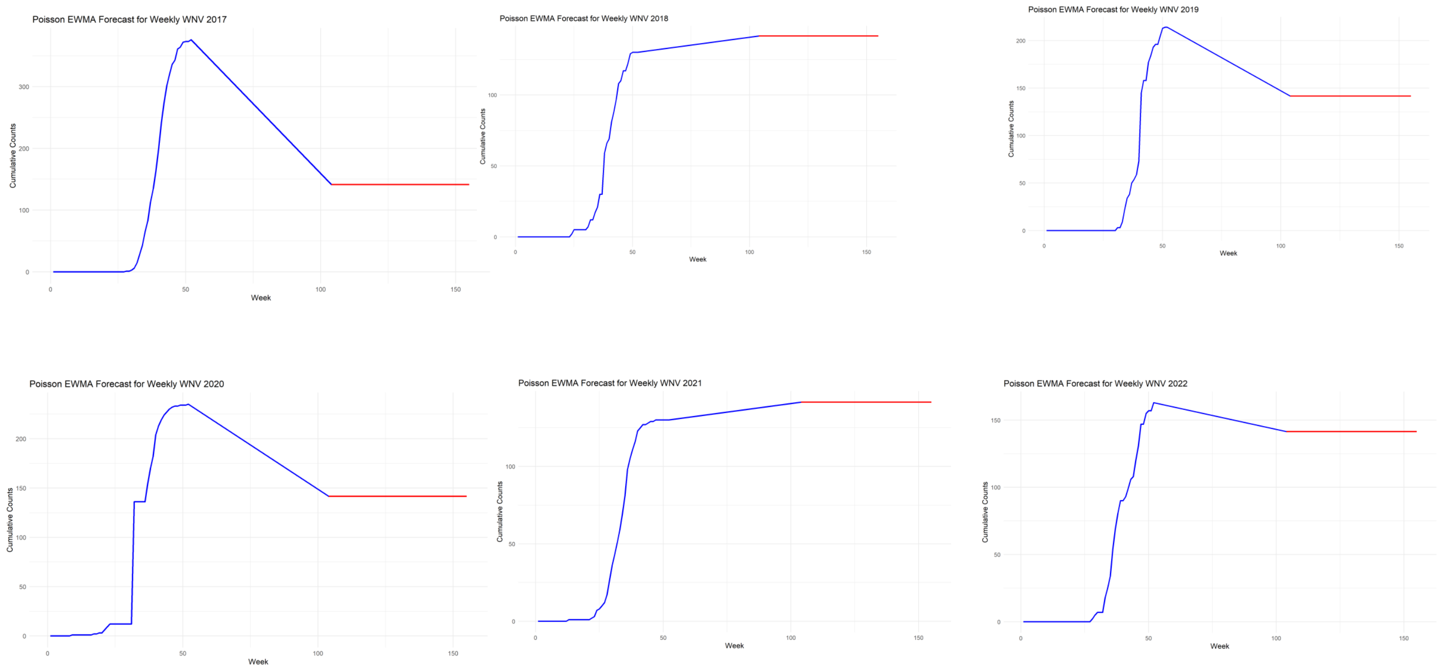
2017 Monthly ACF of Residuals- Poisson EWMA
2018 Monthly ACF of Residuals- Poisson EWMA
2019 Monthly ACF of Residuals- Poisson EWMA
2020 Monthly ACF of Residuals- Poisson EWMA
2021 Monthly ACF of Residuals- Poisson EWMA
2022 Monthly ACF of Residuals- Poisson EWMA
ACF
Lag

Figure 12



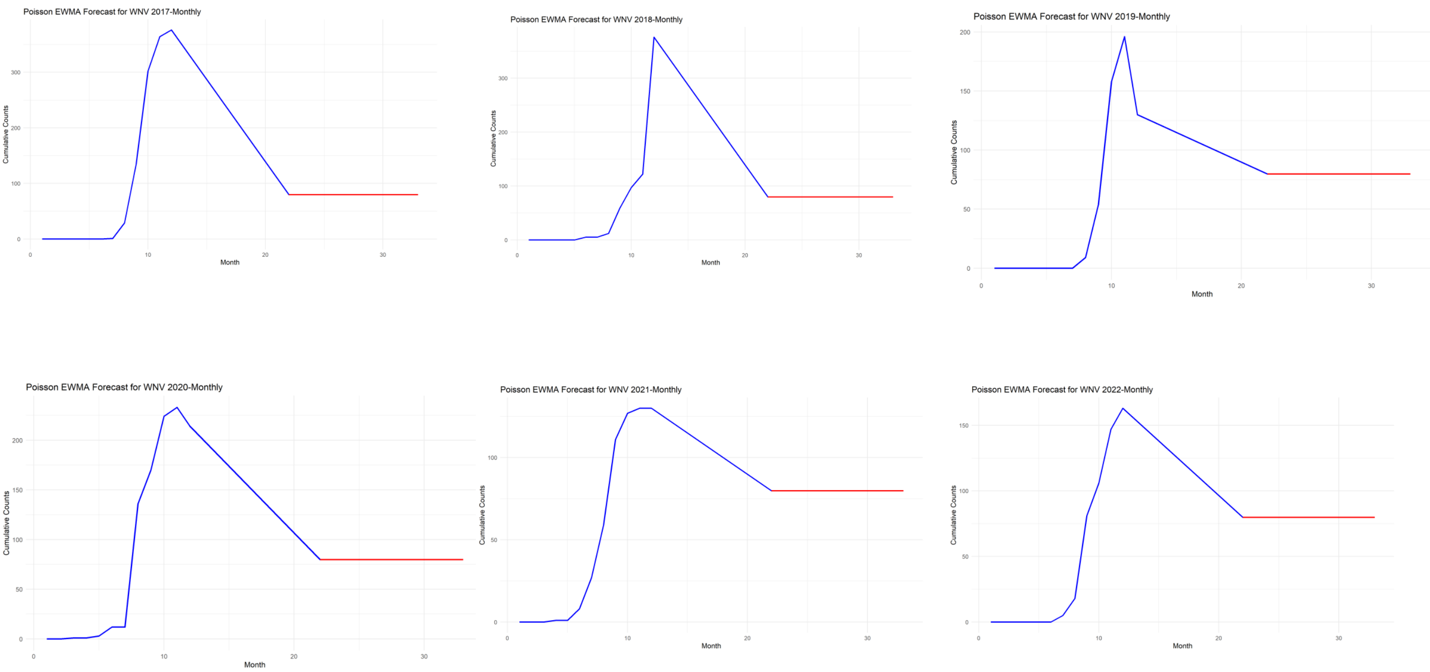
SARIMA ACF of Monthly WNV 2017 Cumulative Counts
SARIMA ACF of Monthly WNV 2018 Cumulative Counts
SARIMA ACF of Monthly WNV 2019 Cumulative Counts
SARIMA ACF of Monthly WNV 2020 Cumulative Counts
SARIMA ACF of Monthly WNV 2021 Cumulative Counts
SARIMA ACF of Monthly WNV 2022 Cumulative Counts
ACF
Lag

Figure 13



Poisson EWMA Forecast for Weekly WNV 2017
Cumulative Counts
Week
Poisson EWMA Forecast for Weekly WNV 2018
Cumulative Counts
Week
Poisson EWMA Forecast for Weekly WNV 2019
Cumulative Counts
Week
Poisson EWMA Forecast for Weekly WNV 2020
Cumulative Counts
Week
Poisson EWMA Forecast for Weekly WNV 2021
Cumulative Counts
Week
Poisson EWMA Forecast for Weekly WNV 2022
Cumulative Counts
Week

Figure 14


Poisson EWMA Forecast for WNV 2017-Monthly
Cumulative Counts
Month
Poisson EWMA Forecast for WNV 2018-Monthly
Cumulative Counts
Month
Poisson EWMA Forecast for WNV 2019-Monthly
Cumulative Counts
Month
Poisson EWMA Forecast for WNV 2020-Monthly
Cumulative Counts
Month
Poisson EWMA Forecast for WNV 2021-Monthly
Cumulative Counts
Month
Poisson EWMA Forecast for WNV 2022-Monthly
Cumulative Counts
Month

Discussion

This study investigates the complex interplay between West Nile Virus (WNV) outbreaks, air quality indicators such as $PM_{2.5}$, and various climatic factors over a defined period. The findings suggest that environmental conditions, particularly air pollution, may significantly influence the transmission dynamics of WNV. While our analyses using Generalized Additive Models (GAM) and Causal Deep Learning (CDL) did not establish a direct causal relationship between $PM_{2.5}$ or AQI levels and neuroinvasive cases, they highlight the intricate and multifaceted nature of these interactions.

Our findings from the GAM analysis reveal that while there are observable trends in the association between PM2.5 levels and Neuroinvasive cases, the smoothed terms s(PM2.5) and s(AQI) did not exhibit statistically significant causal relationships with variations in Neuroinvasive outcomes. This suggests that while air pollution may play a role in public health outcomes, other environmental or biological factors are likely contributing to WNV's transmission dynamics. Future research should aim to identify these additional variables, potentially including other pollutants, climate variables, and ecological factors that influence mosquito populations and their ability to transmit the virus.

Furthermore, the use of CDL allowed us to simulate policy impacts regarding $PM_{2.5}$ reductions. By simulating a 10 % reduction in PM2.5 levels, we observed a potential decrease in predicted Neuroinvasive cases. The ability to predict outcomes based on counterfactual scenarios provides valuable insights for policymakers, emphasizing the importance of air quality regulations in mitigating public health risks associated with WNV.

The seasonal analysis presented in Figures 4 to 7 underscores a clear seasonal trend in WNV cases in California, with significant increases observed during warmer months. This correlation between seasonality in WNV cases and $PM_{2.5}$ concentrations suggests a potential relationship that warrants further investigation. Enhanced monitoring of air quality during high-risk seasons could serve as an early warning system for public health interventions aimed at preventing WNV outbreaks.

Moreover, our application of Poisson Event Weighted Moving Average (PEWMA) and Seasonal Autoregressive Integrated Moving Average (SARIMA) models for forecasting indicates their effectiveness in capturing underlying trends and seasonal patterns. While PEWMA demonstrated superior adaptability in handling noise and short-term fluctuations, SARIMA remains a robust choice for understanding long-term seasonal dynamics. This dual approach enriches our forecasting capabilities, allowing for informed decision-making in public health planning.

In light of the significant resurgence of WNV in Colorado during 2023, it is imperative to understand the factors driving these outbreaks. The observed relationship between air quality and WNV suggests that addressing pollution levels could mitigate the impacts of this mosquito-borne virus. Therefore, future public health policies should focus on reducing $PM_{2.5}$ emissions and enhancing air quality as part of broader strategies to combat WNV transmission.

The limitation is that exploring the county-level West Nile virus and $PM_{2.5}$ data for each state would be more meaningful. Still, there is currently no weekly county-level West Nile virus data available.Understanding the local dynamics of West Nile virus transmission about $PM_{2.5}$ levels would provide valuable insights into potential correlations and health impact. Moreover, without granular data, we are unable to identify patterns or trends that could inform public health interventions tailored to specific regions. The interplay between environmental factors like air quality and vector-borne diseases is complex, and county-level data would allow for a more nuanced exploration of how $PM_{2.5}$ levels might influence West Nile virus prevalence. Additionally, this limitation underscores the need for improved data collection efforts at the county level Without such data, researchers may have to rely on aggregated state-level statistics, which may mask important local variations and lead to less effective strategies for mitigating the risks associated with West Nile virus.

Conclusion

In conclusion, this study provides critical insights into the relationship between West Nile Virus transmission, air quality, and climatic factors. While direct causality between $PM_{2.5}$ levels and neuroinvasive cases remains ambiguous, the observed trends underscore the need for comprehensive public health strategies that address environmental health as a component of infectious disease management. The methodologies employed, including GAM and CDL, offer innovative avenues for predicting and understanding the dynamics of WNV, highlighting the potential for data-driven policy interventions.

As WNV continues to pose a public health threat, particularly in regions with fluctuating climate conditions and rising pollution levels, it is essential to prioritize further research into the interconnectedness of environmental factors and infectious diseases. By integrating air quality monitoring with WNV surveillance, public health officials can better prepare for and respond to potential outbreaks, ultimately safeguarding community health.

The findings of this research advocate for actionable strategies aimed at reducing air pollution and improving public health outcomes, particularly in light of climate change's implications for both environmental and public health. Continued interdisciplinary research will be crucial in developing targeted interventions that not only address the symptoms of infectious diseases like WNV but also their underlying environmental causes.

Acknowledgement

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Data and code availability statement

AQI data from `https://aqs.epa.gov/aqsweb/airdata/download_files.html#AQI`, PM2.5 data from `https://www.epa.gov/outdoor-air-quality-data/download-daily-data` and `https://sites.wustl.edu/acag/datasets/surface-pm2-5/`.

The code is made available from Github repository.

Conflicts of Interest

The authors declare no conflict of interest.

Appendix A: Environment

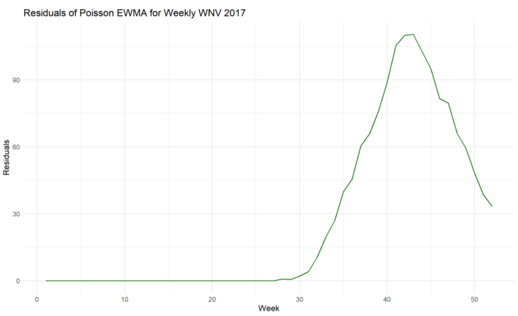
We use Python version 3.11.5, packaged by Anaconda, Inc. ,(main, Sep 11, 2023, 13:26:23) [MSC v.1916 64 bit (AMD64)], with default parameter. The R version is 4.3.1 (2023-06-16 ucrt).

The R version information is as follows: the platform is `x86_64-w64-mingw32`, with an architecture of `x86_64` and operating system `mingw32`. The C runtime is `ucrt`, and the system is described as `x86_64, mingw32`. The version details are: major version 4, minor version 3.1, released on June 16, 2023. The SVN revision is 84548. The version string is `R version 4.3.1 (2023-06-16 ucrt)`. The nickname for this version is *Beagle Scouts*.

Appendix B: Model diagnostics and setting

For the selection and diagnosis of the ARIMA model, Cao et al. provide a detailed introduction (Cao, 2024b). For the PEWMA model, we first extract the columns related to the cumulative counts of West Nile Virus (WNV) for the year 2022 from the dataset, ensuring that the date format is correct. Next, we define a function to calculate the Poisson Exponentially Weighted Moving Average (EWMA) and set the smoothing parameter λ to compute the EWMA values for WNV. We then check for any NA values and remove any rows containing NA values, if necessary, to ensure data integrity. Afterward, we recalculate the EWMA and residuals, and plot the residuals to perform model diagnostics, checking for autocorrelation and normality of the residuals. Finally, we define a forecasting function to predict the next 12 months based on the latest EWMA values and plot a comparison of the original data with the forecasted data to demonstrate the model's effectiveness and trends. These steps will aid in analyzing the time series data of WNV counts and provide effective predictions for incidence rates under varying PM2.5 conditions.

For the SARIMA model, we first convert the cumulative count data of WNV for 2022 into a time series object and plot the time series graph to identify seasonality. Next, we use the ACF and PACF plots to determine the parameters for the SARIMA model, and then fit the SARIMA model using the 'auto.arima' function and review the model summary. Afterward, we conduct forecasts for the next 12 months and plot the forecast results. Finally, we evaluate the model's performance using AIC, BIC, and residual analysis to ensure that the model effectively captures the seasonal patterns in the data.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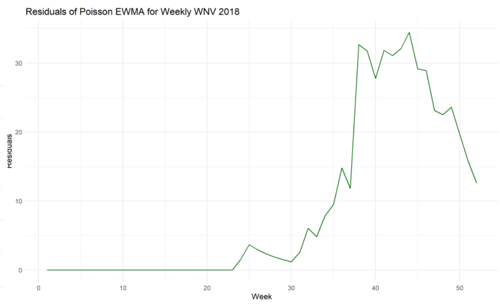
Residuals
Week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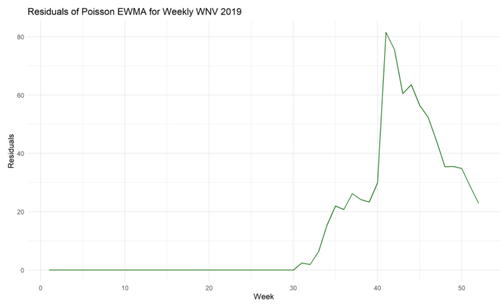
Residuals
Week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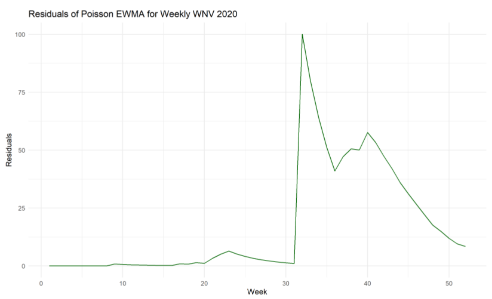
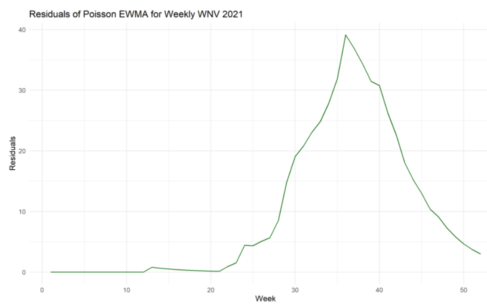
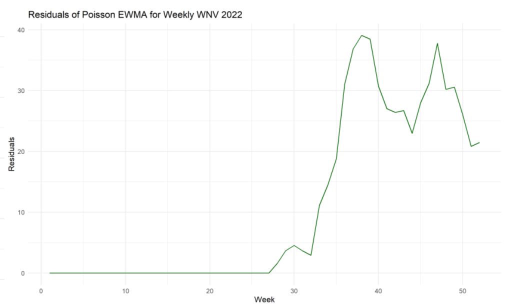
Residuals
Week


Residuals of Poisson EWMA for Weekly WNV 2020
Residuals
Week


Residuals of Poisson EWMA for Weekly WNV 2021
Residuals
Week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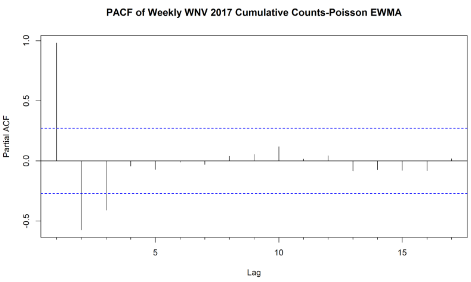
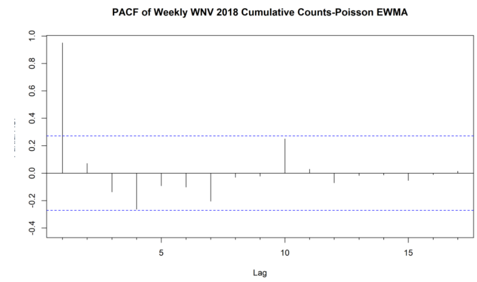
Residuals
Week


PACF of Weekly WNV 2017 Cumulative Counts-Poisson EWMA
Partial ACF
Lag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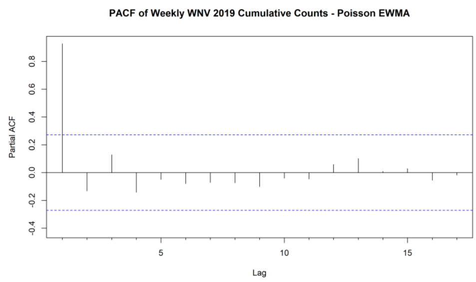
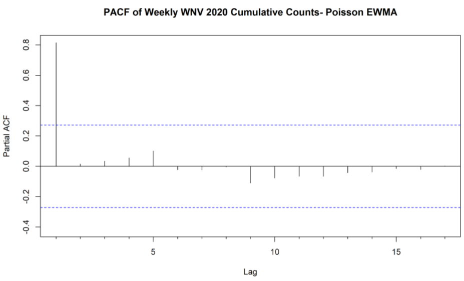
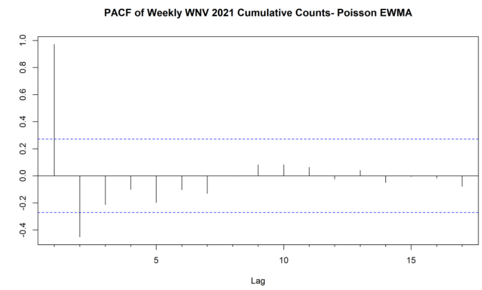
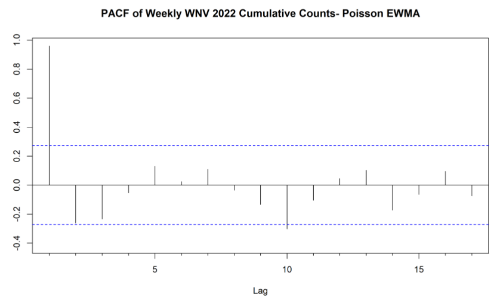
Lag


PACF of Weekly WNV 2019 Cumulative Counts - Poisson EWMA
Partial ACF
Lag


PACF of Weekly WNV 2020 Cumulative Counts- Poisson EWMA
Partial ACF
Lag


PACF of Weekly WNV 2021 Cumulative Counts- Poisson EWMA
Lag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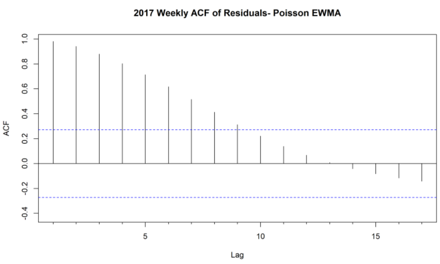
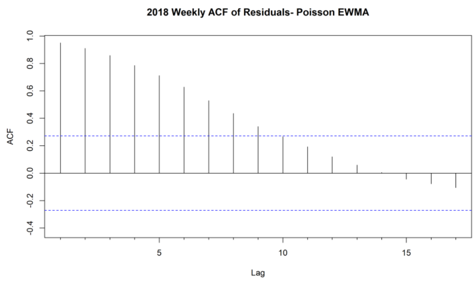

2017 Weekly ACF of Residuals- Poisson EWMA
ACF
Lag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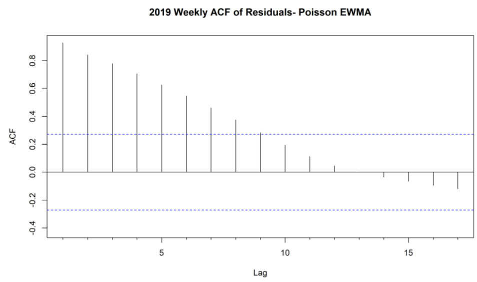
ACF
Lag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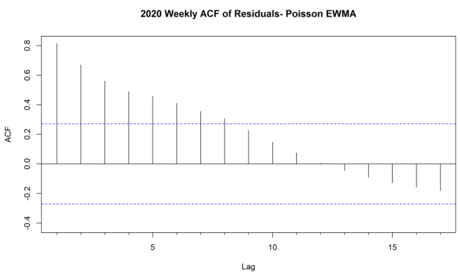
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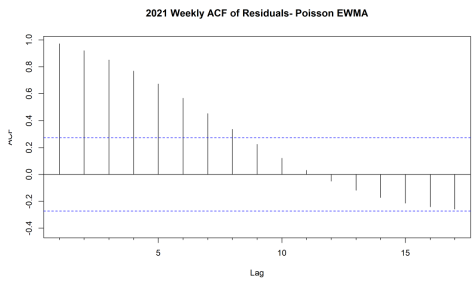
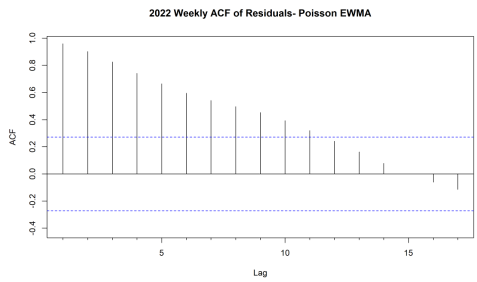
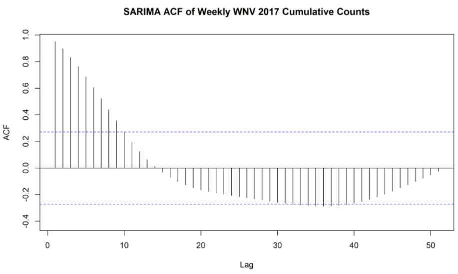
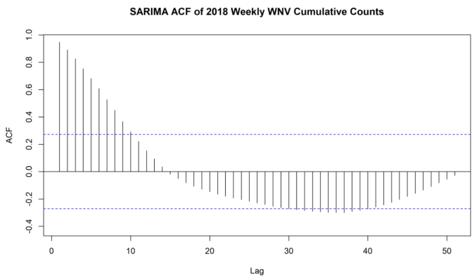

2021 Weekly ACF of Residuals- Poisson EWMA
ACF
Lag


2022 Weekly ACF of Residuals- Poisson EWMA
ACF
Lag


SARIMA ACF of Weekly WNV 2017 Cumulative Counts
ACF
Lag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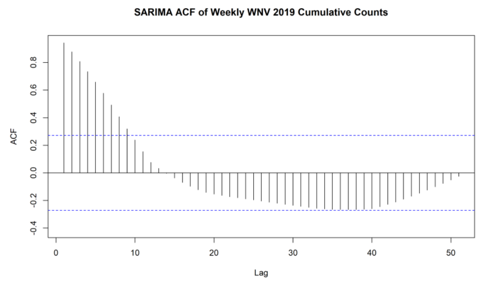
Lag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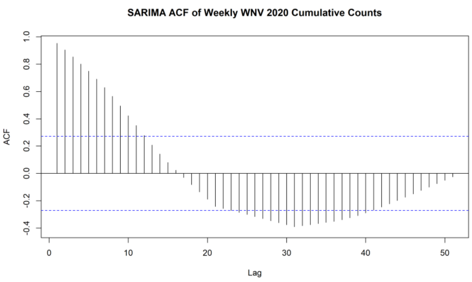
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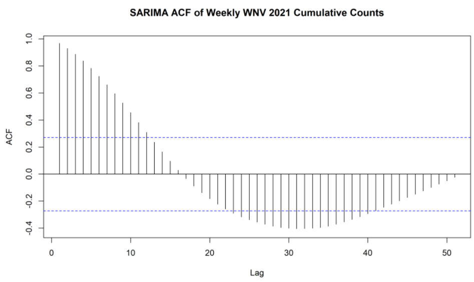
Lag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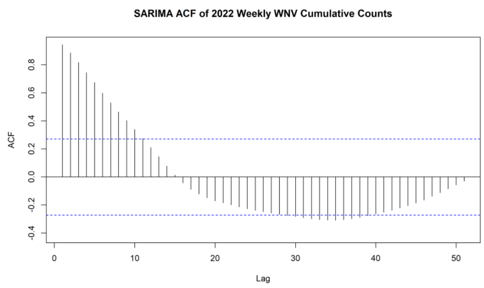
Lag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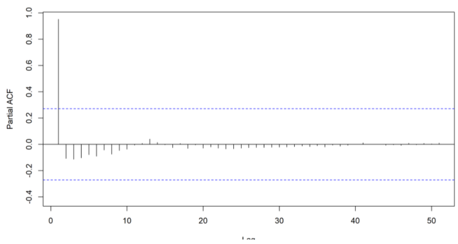
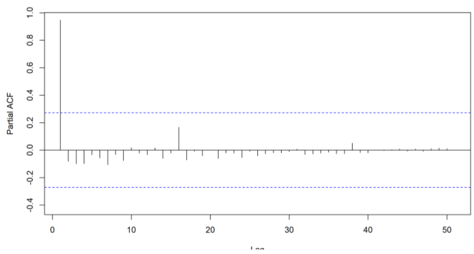
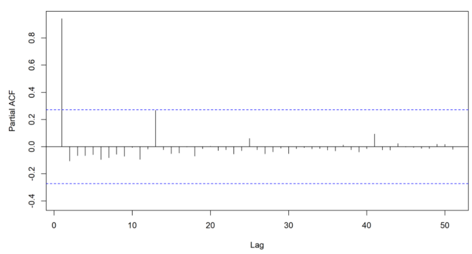
SARIMA PACF of Weekly WNV 2017 Cumulative Counts



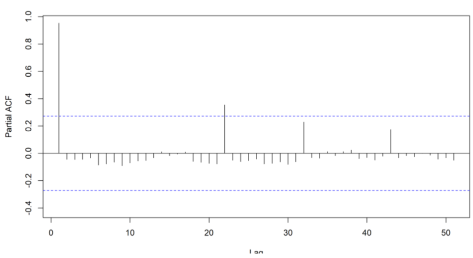
SARIMA PACF of Weekly WNV 2018 Cumulative Counts



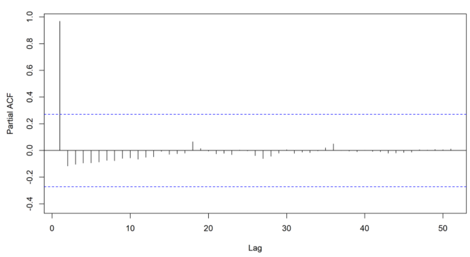
SARIMA PACF of Weekly WNV 2019 Cumulative Counts



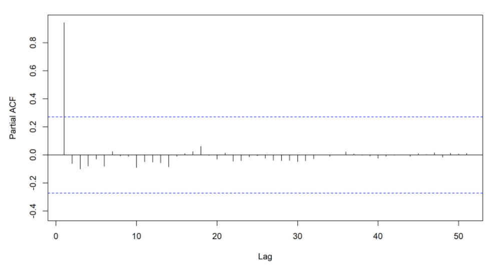
SARIMA PACF of Weekly WNV 2020 Cumulative Counts

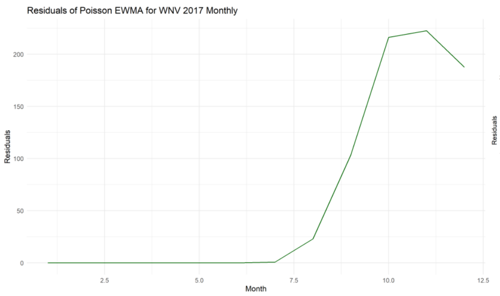


SARIMA PACF of Weekly WNV 2021 Cumulative Counts



SARIMA PACF of 2022 Weekly WNV Cumulative Counts




Residuals of Poisson EWMA for WNV 2017 Monthly
Residuals
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150
200
Month
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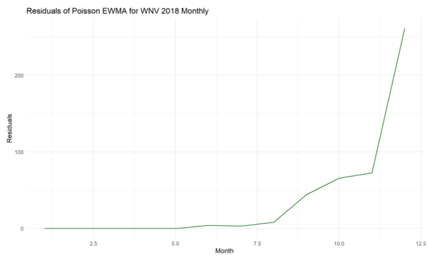
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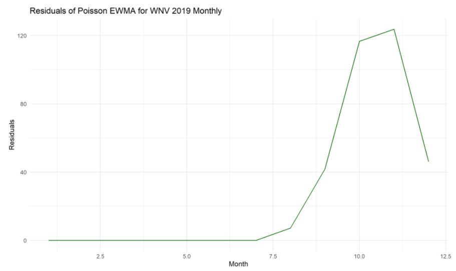
10.0
12.5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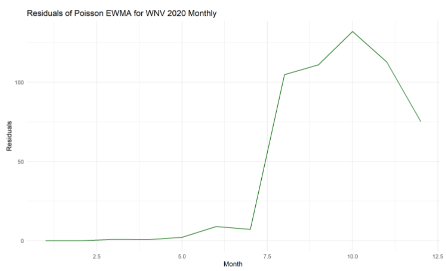
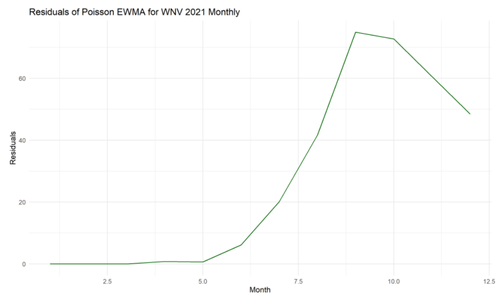
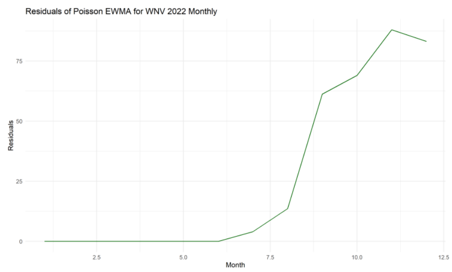
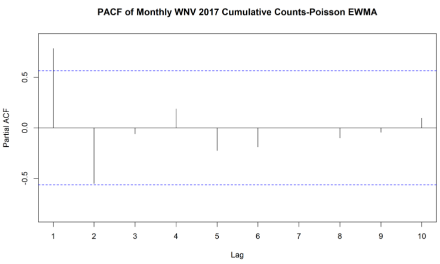
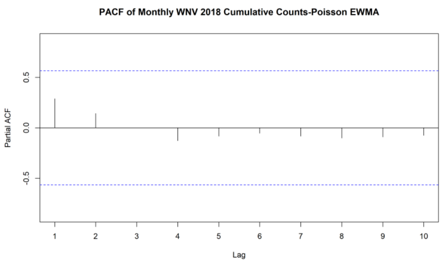
10.0
12.5

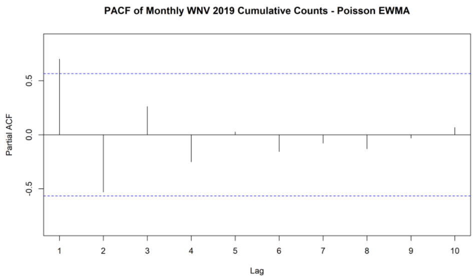

Residuals of Poisson EWMA for WNV 2020 Monthly
Residuals
0
50
100
Month
2.5
5.0
7.5
10.0
12.5

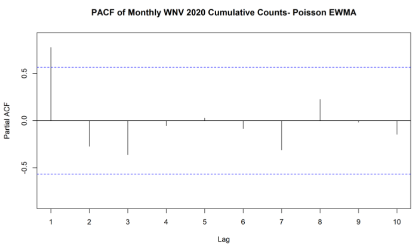

Residuals of Poisson EWMA for WNV 2021 Monthly
Residuals
0
20
40
60
Month
2.5
5.0
7.5
10.0
12.5

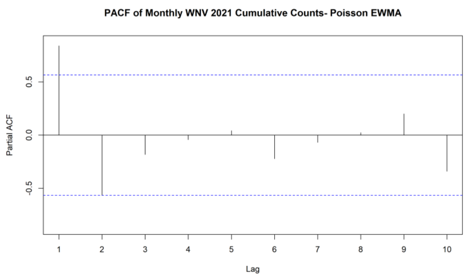

Residuals of Poisson EWMA for WNV 2022 Monthly
Residuals
0
25
50
75
Month
2.5
5.0
7.5
10.0
12.5

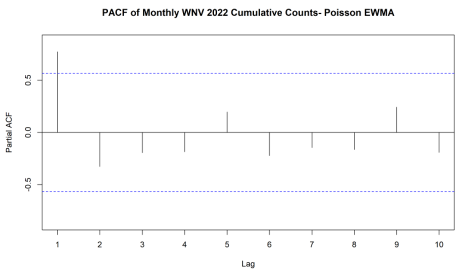

PACF of Monthly WNV 2017 Cumulative Counts-Poisson EWMA
Partial ACF
Lag

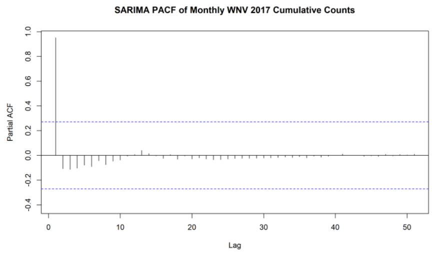

PACF of Monthly WNV 2018 Cumulative Counts-Poisson EWMA
Partial ACF
Lag

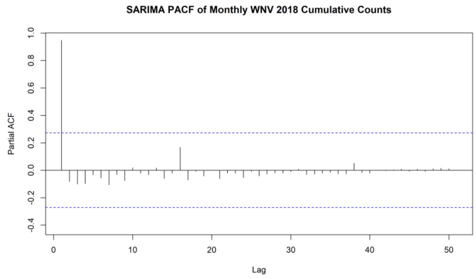

PACF of Monthly WNV 2019 Cumulative Counts - Poisson EWMA
Partial ACF
Lag

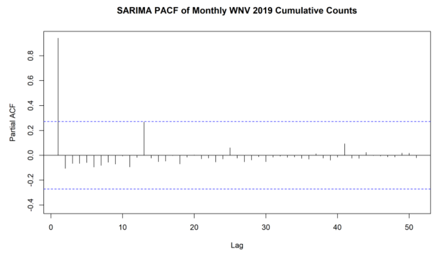

PACF of Monthly WNV 2020 Cumulative Counts- Poisson EWMA
Partial ACF
Lag

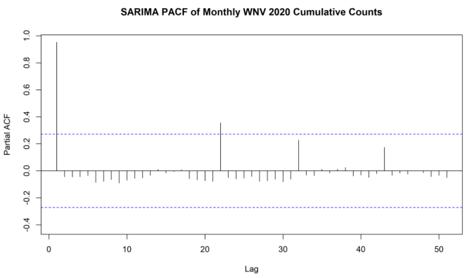

PACF of Monthly WNV 2021 Cumulative Counts- Poisson EWMA
Partial ACF
Lag

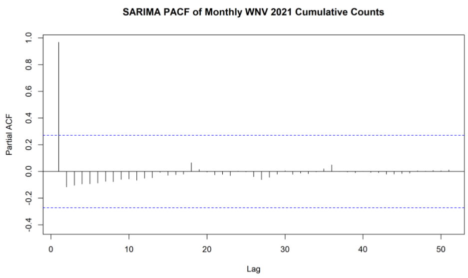

PACF of Monthly WNV 2022 Cumulative Counts- Poisson EWMA
Partial ACF
Lag

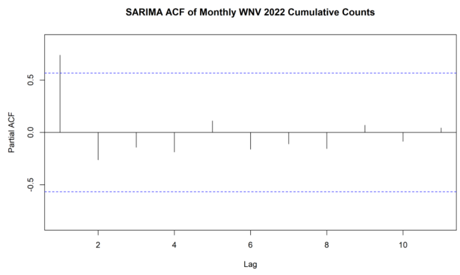

SARIMA PACF of Monthly WNV 2017 Cumulative Counts
Partial ACF
Lag


SARIMA PACF of Monthly WNV 2018 Cumulative Counts
Partial ACF
Lag


SARIMA PACF of Monthly WNV 2019 Cumulative Counts
Partial ACF
Lag


SARIMA PACF of Monthly WNV 2020 Cumulative Counts
Partial ACF
Lag


SARIMA PACF of Monthly WNV 2021 Cumulative Counts
Partial ACF
Lag


SARIMA ACF of Monthly WNV 2022 Cumulative Counts
Partial ACF
Lag

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